

EE561: POWER ELECTRONICS LABORATORY

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Experiment 9: Output voltage Regulation of Buck Converter using Type-II Compensator

❖ Objective

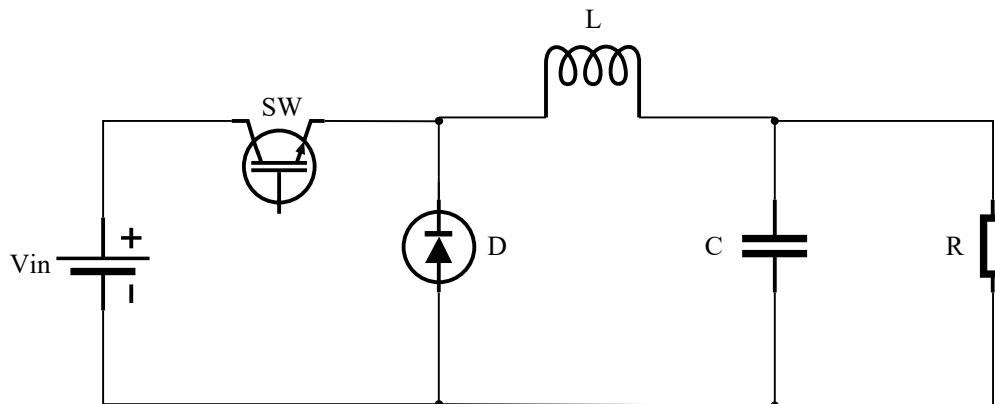
The objective of this experiment is to design a Type-II compensator for a buck converter to regulate its output voltage.

❖ Parameters

Parameter	Value
Input voltage V_{in}	24 V
Capacitor C	100 μ F
Switching frequency f_{sw}	100 kHz
Capacitor C	200 μ F
Load Resistance R	2 Ω
Desired Gain cross-over frequency of compensated system	100 Hz
Desired Phase Margin	90 to 120 degrees

Buck Converter

❖ Circuit Diagram



Buck Converter Circuit

❖ Theory

To control the different parameters in a buck converter, the dynamic model of the converter is required. Through dynamic modelling, we derive the transfer function of the converter. The buck converter small signal output voltage v_o depends on input voltage v_{in} and duty cycle d . The transfer functions with respect to duty cycle is given as:

$$\frac{v_o(s)}{d(s)} \Big|_{v_{in}=0} = \frac{V_{in}}{LCs^2 + \frac{LS}{R} + 1}$$

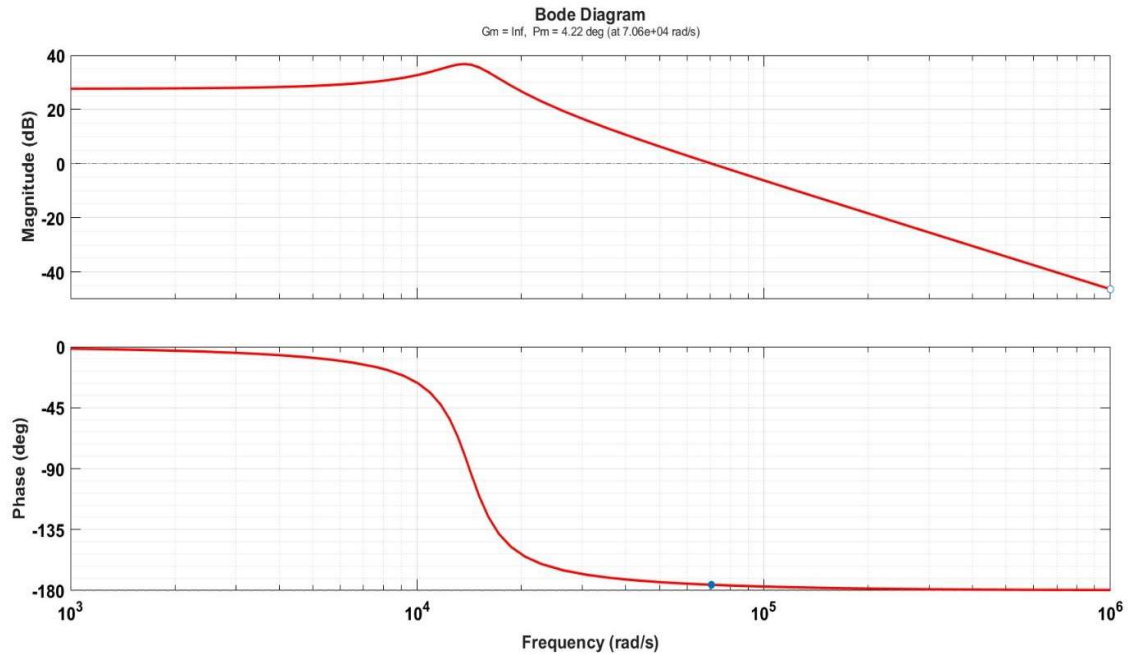
To design a type II compensator for this system, we used k-factor method.

❖ Designing the compensator using k-factor method

1. Get the plant transfer function. From MATLAB script results,

$$G_p(s) = \frac{24}{5 \times 10^{-9}s^2 + 2.5 \times 10^{-5}s + 1}$$

2. Get the bode plot of the plant.



3. Choose the desired gain cross-over frequency. In this case the desired gain-crossover frequency is $f_c = 100 \text{ Hz}$.
4. Choose the desired phase margin. In this case the desired phase margin is $PM = 100^\circ$.
5. Calculate the required mid-band gain.

$$G_{MB} = \frac{1}{\text{gain of } G_p \text{ at } f_c}$$

where G_p is the plant transfer function. From the bode plot, $|G_p|_{w_c} = 24$.

$$G_{MB} = \frac{1}{24}$$

$$G_{MB} = 0.0146$$

6. Calculate the phase boost.

$$\text{Phase boost} = PM - \phi|_{f_c} - 90^\circ$$

where ϕ is the angle of G_p at f_c . From the bode plot, $\phi|_{f_c} = -0.92$.

$$\text{Phase boost} = 100^\circ + 0.92 - 90$$

$$\text{Phase boost} = 50.92^\circ$$

7. Find the value of k.

$$k = \tan \left[45^\circ + \frac{\text{phase boost}}{2} \right]$$

$$k = \tan \left[45^\circ + \frac{50.92^\circ}{2} \right]$$

$$k = 1.211$$

8. Now,

$$w_z = \frac{2\pi f_c}{k}$$

$$w_z = \frac{2\pi \times 100}{1.211}$$

$$w_z = 518.842$$

And,

$$w_p = 2\pi f_c k$$

$$w_p = 2\pi \times 100 \times 1.211$$

$$w_p = 760.893$$

9. The transfer function of the compensator is given by the equation

$$G_{cs}(s) = \frac{G_{MB} \left(1 + \frac{w_z}{s} \right)}{\left(1 + \frac{s}{w_p} \right)}$$

$$G_{cs}(s) = \frac{0.04168s + 21.63}{0.001314s^2 + s}$$

This is the transfer function of the compensator for the plant designed according to the given parameters.

❖ MATLAB commands

11/5/24 11:42 PM D:\MTech\Semester 1\Power El...\sc2 1.m 1 of 1

```
Vin=24;
R=2;
fs=100*10^3;
C=100*10^-6;
L=50*10^-6;
%now the transfer function is given by%
Gp=tf(Vin,[L*C L/R 1])
Gcs=tf([0.04168 0.04168*518.8427],[1/760.8937 1 0])
Cm=Gp*Gcs
figure(1)
bode(Gp)
margin(Gp)
figure(2)
bode(Gcs)
margin(Gcs)
figure(3)
bode(Cm)
margin(Cm)
```

❖ MATLAB Command Results

MATLAB Command Window

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```
>> sc2_1
```

```
Gp =
```

```

          24
-----
5e-09 s^2 + 2.5e-05 s + 1
```

```
Continuous-time transfer function.
Model Properties
```

```
Gcs =
```

```

0.04168 s + 21.63
-----
0.001314 s^2 + s
```

```
Continuous-time transfer function.
Model Properties
```

```
Cm =
```

```

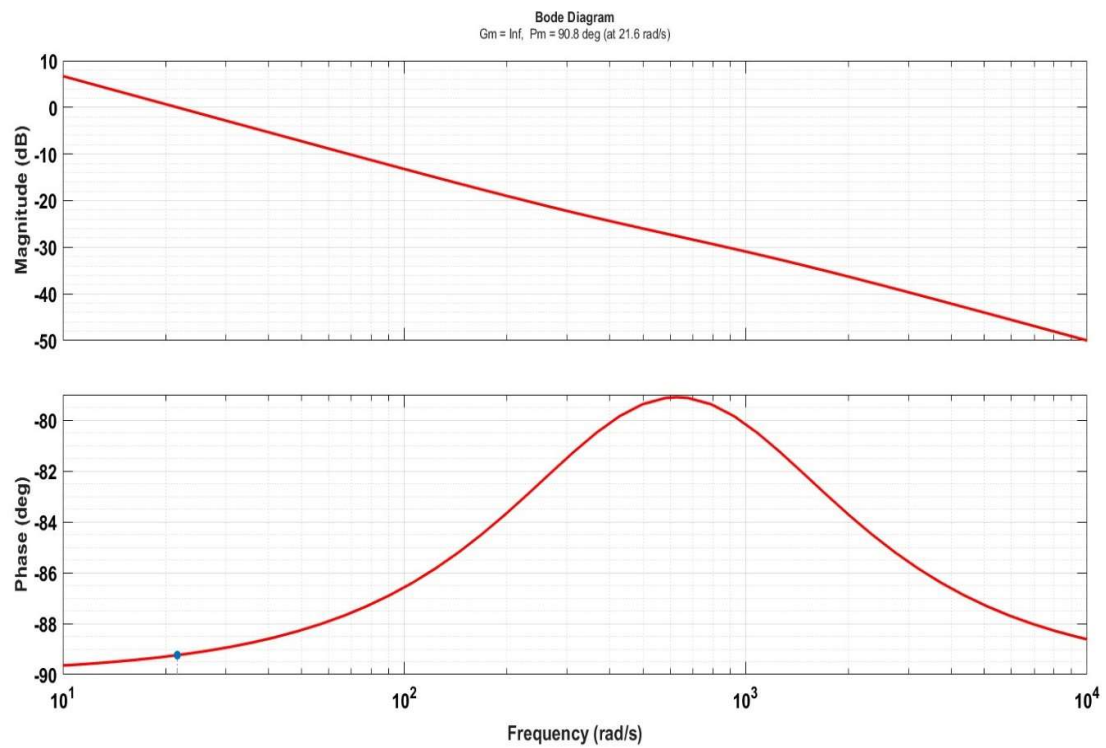
          s + 519
-----
6.571e-12 s^4 + 3.786e-08 s^3 + 0.001339 s^2 + s
```

```
Continuous-time transfer function.
Model Properties
```

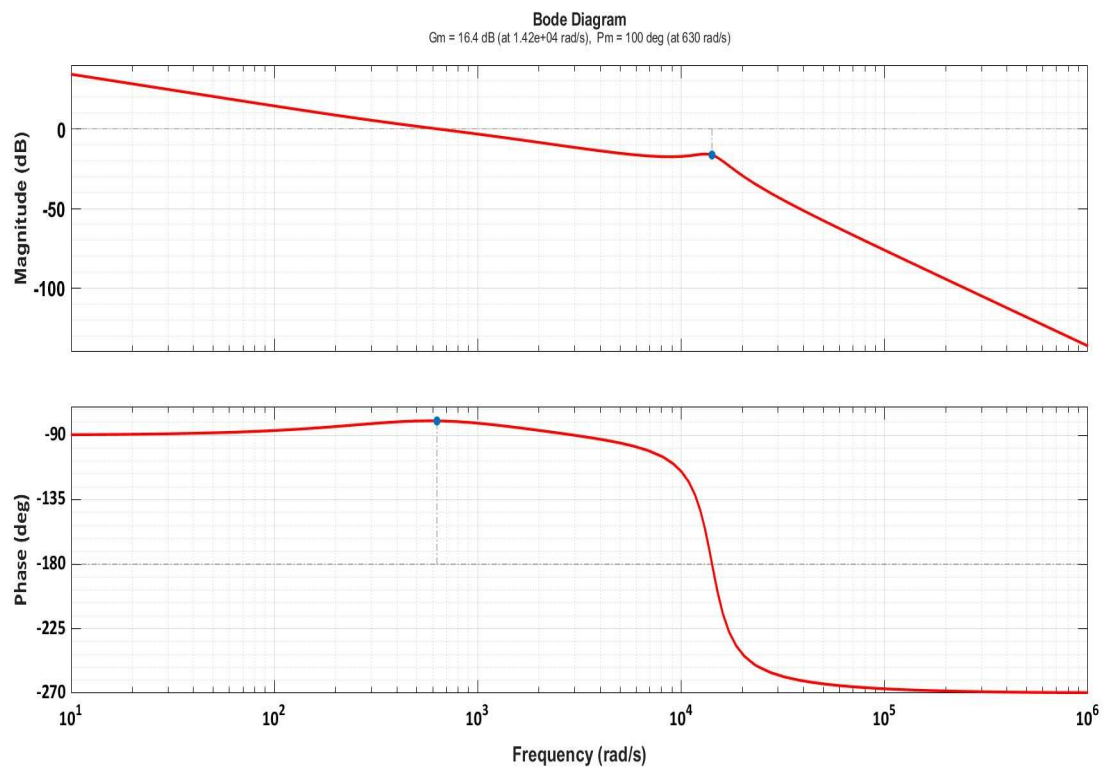
In the results shown above, $C_m(s) = G_p(s)G_{cs}(s)$.

❖ Bode plots

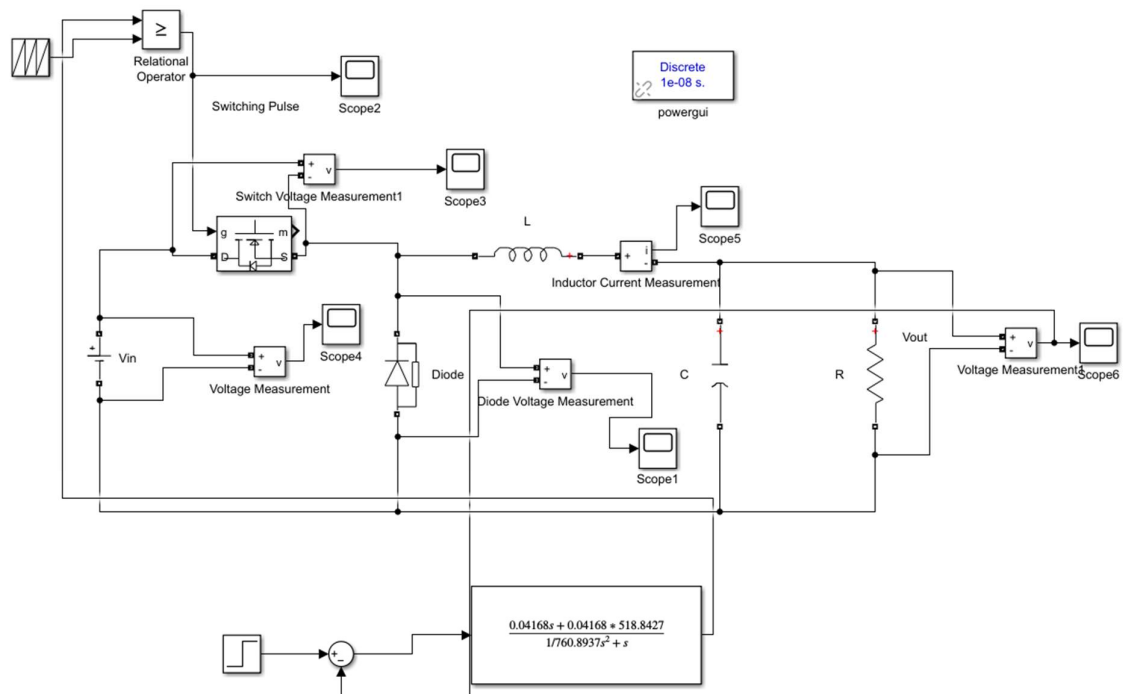
➤ $G_{cs}(s)$



➤ $C_m(s)$

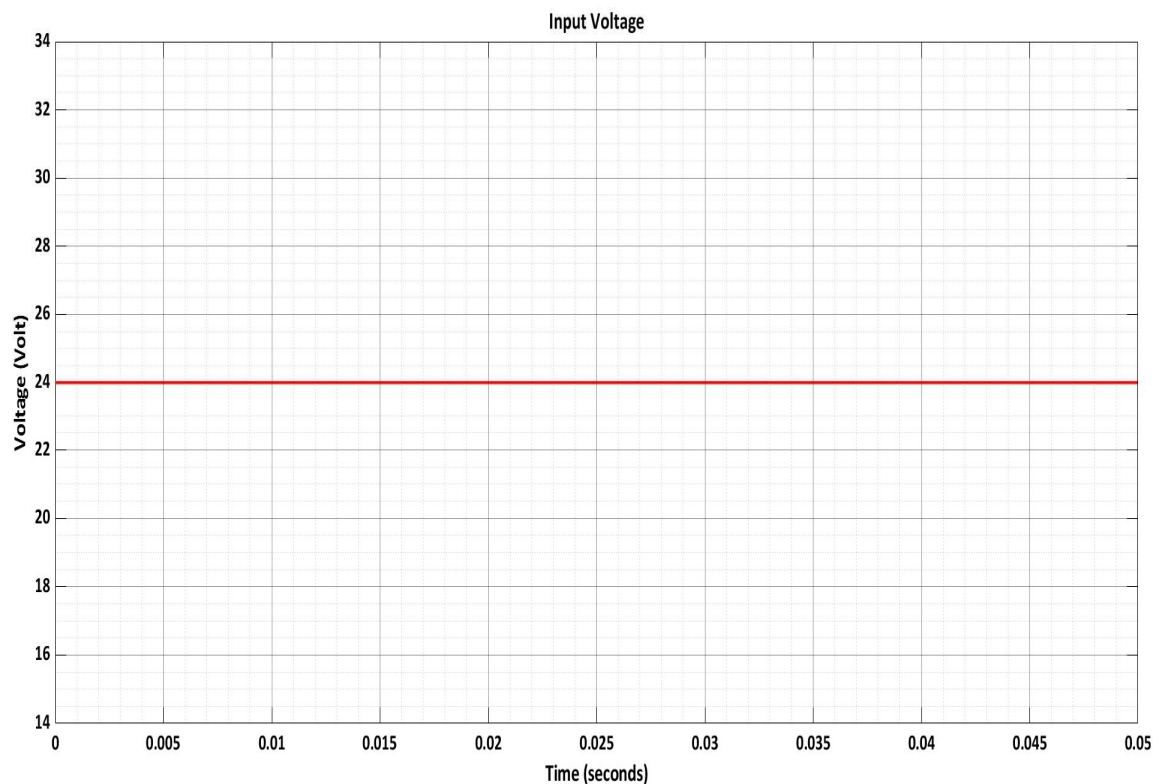


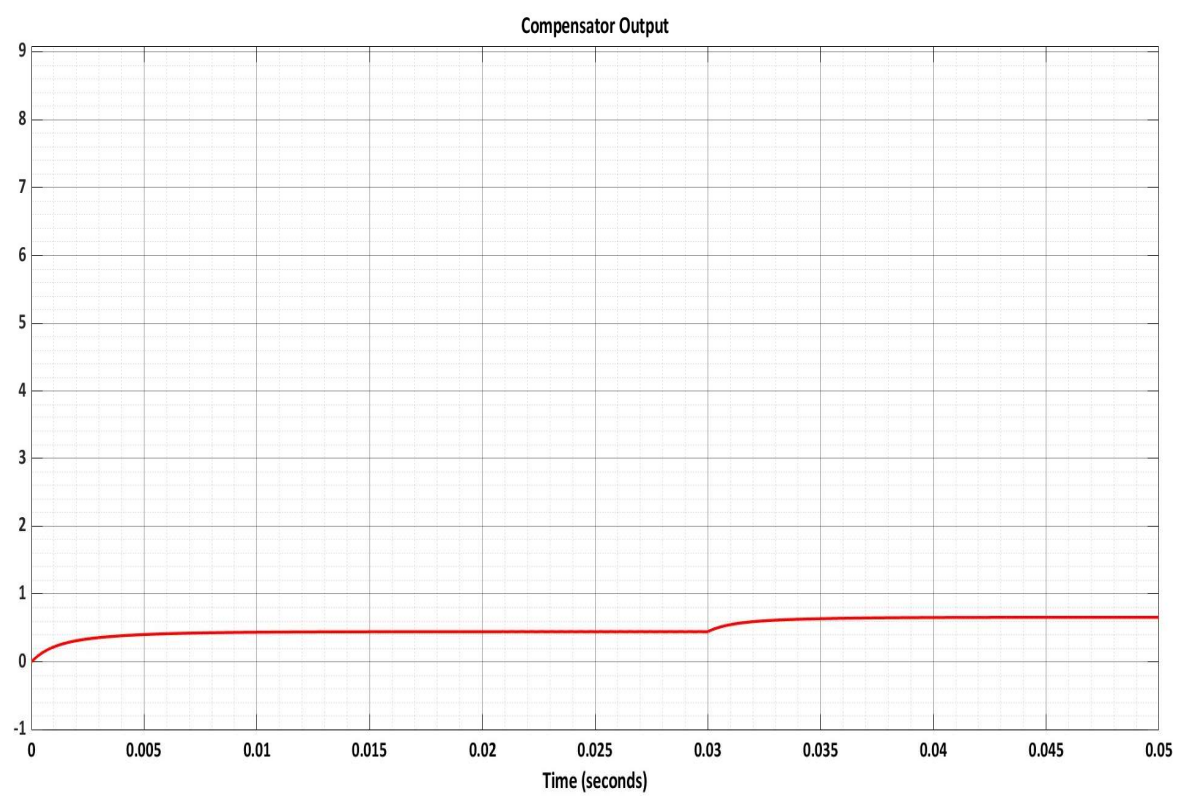
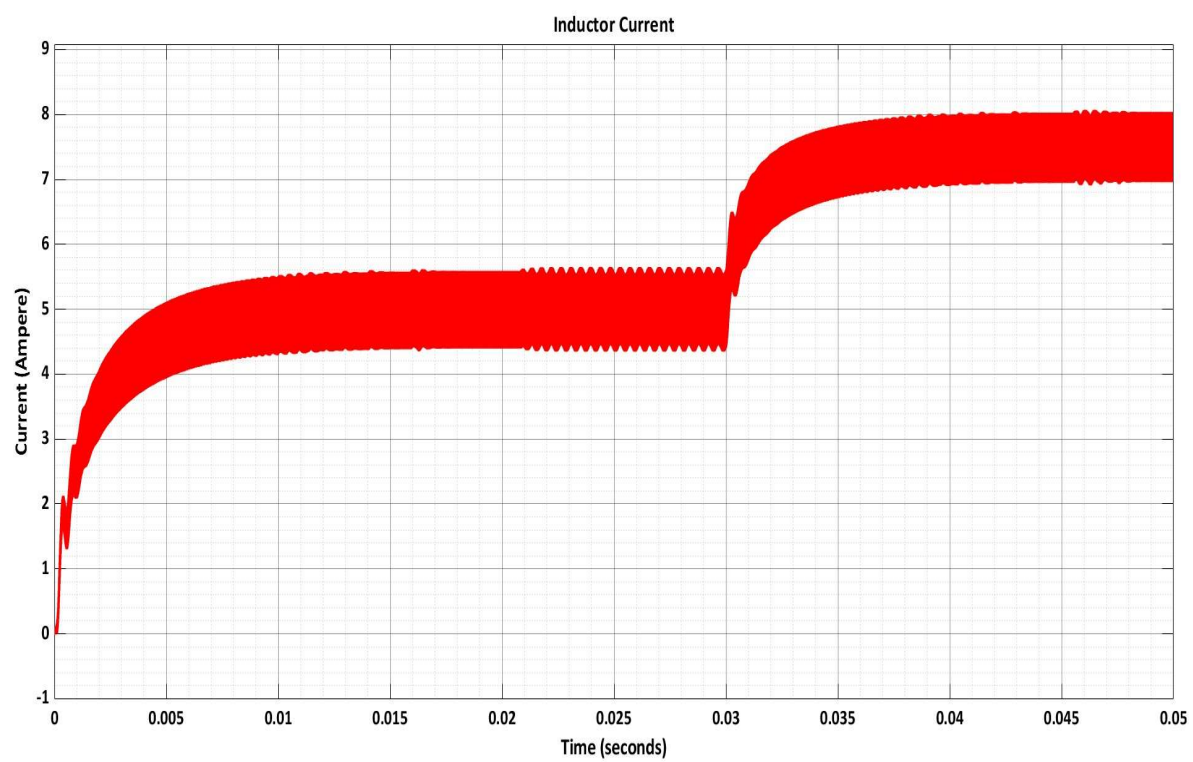
❖ MATLAB/SIMULINK simulation

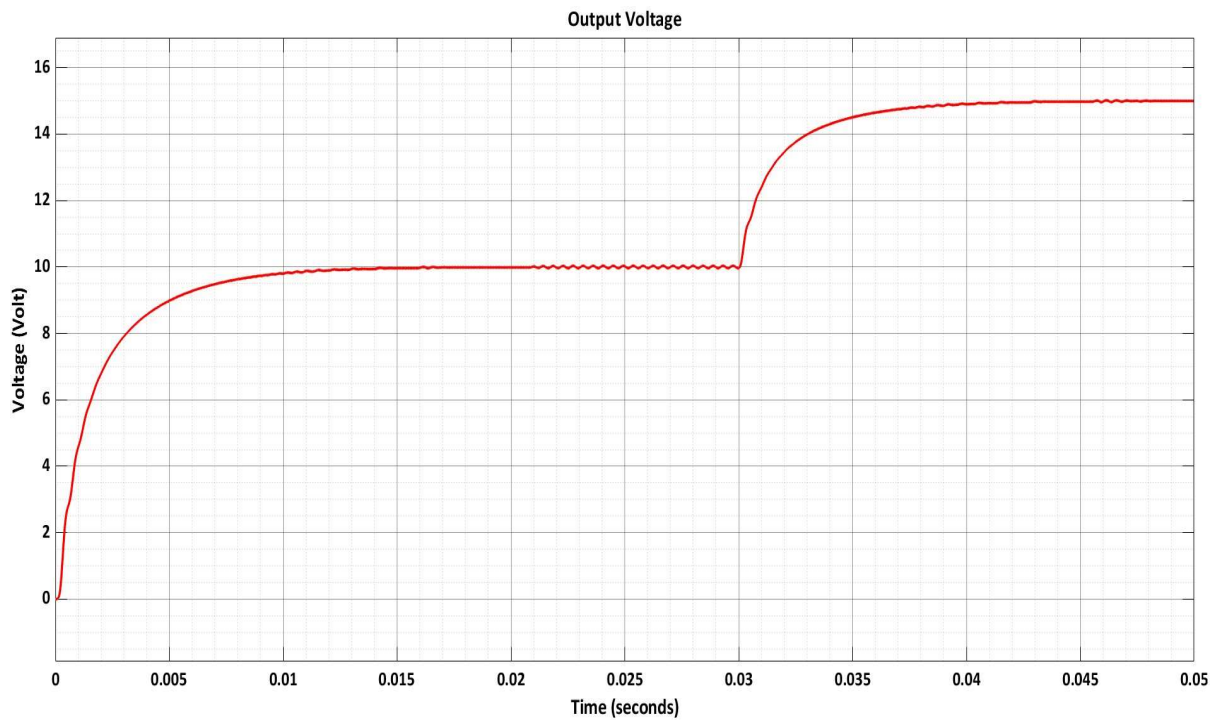


We checked the step response of the above converter by employing the compensator derived above. The buck converter given above has the transfer function $G_p(s)$.

❖ Waveform







The reference voltage to the control system was changed from 10 V to 15 V at $t = 0.03 \text{ sec}$.

❖ Simulation Parameters

